TERTIARY INFOTECH ACADEMY

LAB 04
Deploy and Test a Multi-AZ Highly Available Service

AWS Certified SysOps Administrator Associate Training
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Scenario
Northstar Retail's single-instance web service cannot survive an instance or Availability Zone disruption.

Objective
Apply K4, K5, A3 and A5 by deploying an ALB and Auto Scaling service across two Availability Zones, testing health-based recovery and recording a controlled change.

Deliverable
A two-AZ VPC web tier, ALB, target group, launch template, Auto Scaling group, failover evidence and cleanup record.

Services and tools
VPC · EC2 · ALB · Auto Scaling · CloudWatch · CloudFormation

Safety and cost boundary
- Use only the trainer-authorised learner account and Region.
- Replace placeholders and tag learner-owned resources.
- Capture evidence before cleanup.
- Verify deletion and report retained chargeable resources.

Detailed procedure

Step 1 — Review `ha-web-stack.yaml` and the architecture JPEG. Identify the two failure domains, health-check path, minimum capacity and rollback boundary.
```
open EXPECTED-ARCHITECTURE.jpg
```

Evidence note: _____

Step 2 — Validate the template and deploy it with a unique suffix.
```
aws cloudformation validate-template --template-body file://ha-web-stack.yaml
aws cloudformation deploy --stack-name SysOps-Lab04-REPLACE_SUFFIX --template-file ha-web-stack.yaml --capabilities CAPABILITY_NAMED_IAM
```

Evidence note: _____

Step 3 — Read the stack outputs; save the ALB URL, target group and Auto Scaling group identifiers.
```
aws cloudformation describe-stacks --stack-name SysOps-Lab04-REPLACE_SUFFIX --query 'Stacks[0].Outputs' --output table | tee evidence/stack-outputs.txt
```

Evidence note: _____

Step 4 — Open the ALB URL repeatedly and confirm both Availability Zones contain healthy targets.
```
aws elbv2 describe-target-health --target-group-arn REPLACE_TARGET_GROUP_ARN --output table | tee evidence/target-health-before.txt
```

Evidence note: _____

Step 5 — Inspect desired, minimum and maximum capacity. Confirm the group is configured across two subnets.

```
aws autoscaling describe-auto-scaling-groups --auto-scaling-group-names REPLACE_ASG_NAME --
output json | tee evidence/asg-before.json
```

Evidence note: _____

Step 6 — Run the bounded failure test by terminating one learner-owned instance in the Auto Scaling group; record the UTC start time and change ID.

```
aws ec2 terminate-instances --instance-ids REPLACE_CANARY_INSTANCE_ID | tee
evidence/termination.json
```

Evidence note: _____

Step 7 — Observe target health, replacement activity and ALB availability until desired capacity and health recover.

```
bash failover-observe.sh REPLACE_ASG_NAME REPLACE_TARGET_GROUP_ARN | tee evidence/failover-
observation.txt
```

Evidence note: _____

Step 8 — Calculate detection and recovery time in `FAILOVER-REPORT.md`; explain whether the observed result meets the stated objective.

```
open FAILOVER-REPORT.md
```

Evidence note: _____

Step 9 — Create the target-tracking policy from `scaling-policy.json` only if the trainer approves a scaling demonstration.

```
aws autoscaling put-scaling-policy --cli-input-json file://scaling-policy.json | tee
evidence/scaling-policy.json
```

Evidence note: _____

Step 10 — Capture ALB health, Auto Scaling activity and CloudWatch signals as JPEG evidence.

```
open EXPECTED-ARCHITECTURE.jpg
```

Evidence note: _____

Step 11 — Run validation, delete the stack and wait for DELETE_COMPLETE before leaving the lab.

```
bash validate.sh
bash cleanup.sh | tee evidence/cleanup.txt
```

Evidence note: _____

Verification

The service spans two Availability Zones, returns a healthy response during one-instance replacement, recovers desired capacity, records measured recovery time, and validation emits LAB 04 PASS.

Acceptance criteria

- Two distinct Availability Zones
- Minimum and desired capacity at least two

- Health check represents application response
- Failure test affects only learner-owned canary
- Measured recovery and rollback are documented

Evidence checklist

□ Stack outputs

□ Target health before/after

□ Auto Scaling activity

□ ALB response JPEG

□ completed FAILOVER-REPORT.md

□ cleanup.txt

Troubleshooting

- Check identity, Region and resource ID before changing permissions.
- Resolve every REPLACE placeholder.
- Preserve error output as evidence.
- Escalate cleanup failures immediately.